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Natural food colorants: Extraction and stability study

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ABSTRACT

The food colorant market is growing at a 4.6% annual rate, and it is estimated to reach 2.3 billion dollars worth of the global market. The color of the food is considered the most critical aspect while purchasing any food items, as it is most likely associated with the freshness and quality of the product. Synthetic and natural colorants are added to enhance the color properties of a food product. However, synthetic food colorants cause long-term adverse health impacts. Therefore, natural colorant-related topics have gained significant interest in research field. Natural colorants with the highest applicability are anthocyanins, carotenoids, and betacyanins, because of which they are discussed in brief. From the literature review, it has been found that the current research trend is to find different sources, various feasible and effective extraction methods of natural colorants. Even after the effective extraction of colorant, the stability problem arises. As stability is associated with storage time, it is crucial to improve the stability of the extracted matrix. For this purpose, various microencapsulation and nanoencapsulation methods are devised by various researchers. Furthermore, the degradation mechanism of various food pigments is also discussed in brief.
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1. Introduction

The color of any food object is one of the essential aspects representing its freshness and safety, and pleasing aesthetics. Food pigments derived from natural resources have become a global research interest and are continuously growing in importance. As synthetic food pigments have undesirable properties, natural food pigments are used in their substitute because of their positive health effect [1].For a long time, color additives have been used to improve the aesthetic value of foods, drinks, cosmetics, and medications. Demand for a natural source of such compounds is increasing day by day due to the awareness of natural compound's positive health benefits [2].Hence, it is necessary to look for novel sources of natural food pigments. The quality of the food product is often determined by how eye-catching it is, as the buyers of the food product have specific expectations with the color of the product. Also, color has a substantial psychological impact [2]. The consumer inspects the food by its color during the initial inspections. Not only this, whether food will be accepted or rejected also depends upon it. The color of meal has great effect on the stimulation of the appetite. Therefore, food industry makes extra effort on the color and appearance of its product to increase the appetite and temptation before consumption [3]. For example, dried tofu is colored by adding food colors like sunset yellow or soy sauce [4].

The food colorants can be defined as those that impart color to the food that may be natural or synthetic and are not consumed as foods (FSSAI, 2011). Food colorants are of two types: (i) Natural food colorant (ii) Synthetic food colorant. The food colorant market is growing at 4.6% yearly and is estimated to get a 2.3 billion dollar world market. Food is produced and processed in food industries that may be situated far from the consumer market, and foods are needed to be transported to the consumer. In this case, the food may lose its appearance and some properties. To prevent this from happening, food additives are added. The following are the point wise reasons to use the food colorants [5]:

- To prevent the loss of original food appeal
- To enhance the color in food and the consumer will correlate this intensified color with food quality.
- To preserve vitamins vulnerable to flavor and light.
- To maintain the image or character recognized by the food.

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Food colorants are classified into natural and synthetic color:

(i) Natural colorants: Natural dye present in foods, plants, cosmetics, fruit, seeds, algae, or insect extracted by any extraction method. For example, anthocyanins, betacyanins, carotenes, betaxanthins, flavones, and chlorophylls.

(ii) Synthetic colorants: They are formed by chemical synthesis and are not present in natural organic compounds. They are typically soluble in water due to the presence of a sulphonic or carboxylic acid group. FSSAI has permitted synthetic colors like ponceau 4R, carmoisine, erythrosine, tartrazine, sunset Yellow FCF, indigo carmine, and brilliant blue FCF. Also, the permissible limit of using synthetic colors is 100 ppm.

Some review papers in the field of natural colorants were also analyzed. Rodriuez (2016) reviewed natural colorants and pigments mainly focusing on the sources, various extraction methods, the degradation mechanism of anthocyanins, betanins, and chlorophylls. Furthermore, the review paper lacks an in-depth literature compilation of the stability study of food pigments [6]. Renata (2020) compiled literature data for methods used to obtain the food pigments by freeze-drying method [7]. On the other hand, Gengatharan et al. (2015) described various applications of betalains in functional food [8]. None of the review papers available describes the in-depth literature compilation of sources, extraction methods, degradation mechanism, and stability study of carotenoids, anthocyanins, and betanin of the last five year data. The review presented here keeps all these points in check and are described in brief.

As natural colorants are more beneficial to human health and environmentally friendly than synthetic colorants, they are more preferred, and according to applicability in products, carotenoids, anthocyanins, and betanins are the most widely used and broadly discussed here.

2. Carotenoids

2.1. Introduction to carotenoids

Carotenoid, which is exceptionally high in fruits and vegetables, is an essential food component of the human diet. It is composed of a class of natural fat-soluble pigments comprised of eight molecules of five carbon isoprenoids. It is responsible for red, yellow, and orange coloration. Carotenoids are divided into two groups which are carotenes (includes lycopene, α-carotene, and β-carotene) and xanthophylls (lutein, zeaxanthin, and astaxanthin) [9].

2.2. Sources of carotenoids

Apart from recommended sources of carotenoids in FSSAI guidelines, research has been carried out to find out the novel sources of carotenoids like brown seeweeds [10], *Rhodotorula glutinis* yeast [11], carrot juice processing waste [12], tomatoes [13], guava pulp [14], GAC fruit peel, and pulp [15], apricot and shrimp waste [16], orange peel [17], tomato processing waste [18], vegetable waste matrix [19], *Heterochlorella luteoviridis* microalgae [20], *Xanthophyllomyces dendrorhous* yeast [21], carrot peels [22], pumpkin peels [23], corn gluten meal [24], *Rhodotorula glutinis* yeast [25], citrus fruit waste [26], *Dunaliella salina* algue [27], *Chlorella vulgaris* microalgae [28], pomegranate waste [29], *Heterochlorella luteoviridis* [30], red capsicum [31], *Lycium ferocissimum* fruits [32], persimmon [33], etc.

2.3. Extraction methods of carotenoids

The traditional methods of extraction of carotenoids like sonication, homogenization, and soxhlet extraction are mostly replaced by novel extraction techniques, like supercritical fluid extraction [11], enzyme assisted extraction [13], ultrasound-assisted extraction [14], pulsed electric field extraction, ionic liquid assisted ultrasonic extraction [34], natural deep eutectic solvent assisted microwave extraction [16], etc. The research work carried out in the field of extraction of carotenoids is compiled in the Table 1.

Supercritical fluid extraction (SFE) is novel and green technology which mainly employs CO_2 and is known for being fast and efficient method for extraction of natural products from biomass like, vegetables and fruits. The extraction of components takes place in similar fashion as that of conventional solvents, but supercritical fluid has liquid like properties of solubility capacity and density as well as gas like surface tension, diffusion and viscosity power. Because of combined effect of these properties, the extraction takes place in shorter period of time as compared to conventional solvent. Another benefit of SFE is recovery of high purity components without any toxic residue. The disadvantage of SFE is expensive cost of equipment [22]. About 90% (w/w) of total carotenoids were recovered from vegetable waste by SFE with 30 min of extraction time [19]. From carrot peels, SFE recovered 86.1% of carotenoids recovery at 50 min of optimal time [22]. In comparison, lutein recovery from the corn gluten meal was found to be 2.5 times higher by SFE with 10 min of extraction time than extraction using ethanol and chloroform/dichloromethane (2:1) which required 1 h of extraction time [24]. Furthermore, the recovery of carotenoids from SFE from *Dunaliella salina* was almost half (47%) of the solvent extraction yield with extraction time of 90 min [27]. It was reported that ß-carotene, lutein, zeaxanthin, and β-cryptoxanthin extracted from persimmon were 10.56 μg/g, 15.46 μg/g, 16.81 μg/g, and 33.23 μg/g respectively by use of SFE. It was observed that 30 min of SFE produced a better result than the reference soxhlet method [33].

Pulsed electric field(PEF) is a physical technique in which high intensity pulses of few microsecond duration are applied to increase the permeability of the cell membrane. Total carotenoids recovered from *Rhodotorula glutinis* yeast was found to be 250 mg/g when the pulsed electric field technique was used as a pretreatment of biomass. Researchers suggested that the improvement in carotenoids extraction after the pulsed electric field treatment was attributed to PEF-induced autolysis, which leads to disruption in the interaction of carotenoids molecule with other molecule located in cytoplasm, and this causes disintegration of cell wall [11]. Microwave-assisted extraction is less time-consuming and has higher efficiency. Microwave causes cell wall destruction because of which time of extraction decreases and extraction efficacy increases. From carrot juice processing waste, microwave-assisted extraction was utilized to give carotenoids recovery of 77.48% at optimum conditions of 9.39 min, 165 W of microwave power, and 8.06:1 g/g of oil to waste ratio. When the flaxseed oil and carotenoids rich flaxseed oil were compared, the latter had high phenolic content (214.05 μg/g) and high antioxidant activity (inhibition% of DPPH was 70.67) [12]. When the cell wall degrading enzyme was used, the recovery of lycopene from tomatoes was better than the conventional extraction method. Furthermore, mixed enzymic activities were more effective at the complete cell wall breakdown of plant [35]. The enzyme dose of 25 U/g for 180 min gave the maximum total carotenoid recovery of 5.43 mg/kg/U from tomatoes [13]. Enzymatic liquefaction increased the extract yield and total solids from red capsicum by 2.5 fold to threefold compared to unliquified extract [31].

In the ultrasound-assisted extraction, the acoustic cavitations are the driving force, in which the microbubbles are formed and collapsed to give localized high temperature and pressure. This effect results in the disruption of plant tissue and improving the solvent penetration [36]. Carotenoids were recovered from guava pulp, GAC peel, and orange peel using the ultrasound-assisted extraction [14,15,17].

Table 1
Recent advancement in the extraction of carotenoids.

Source (Unit of concentration)	Extraction method	Extraction condition	Carotenoids						Total carotenoids	Reference
			Fucoxanthin	β-Carotene	Lutein	Lycopene	Trans lutein	Total carotenoid recovery (%)		
Orange peel(d)	(f)	42 °C, 35 min, 15 mL/g	–	–	–	–	–	–	1.85	[37]
Mandarin epicarp (d)	(f)	50 °C, 60 min	–	140.7	–	–	–	–	–	[38]
Brown seaweeds(a)	(g)	50 °C, 12 hr	0.625	0.599	0.58	–	–	–	–	[10]
Rhodotorula glutinis yeast(a)	(h)	80 °C, 50 MPa	–	–	–	–	–	–	250	[11]
Carrot juice processing waste (a)	(i)	165 W, 9.39 min,	–	236.35	–	–	–	–	397.1	[12]
Rhodotorula glutinis yeast(b)	(j)	200 kPa, 2 min	–	–	–	–	–	–	25	[39]
Tomatoes(c)	(k)	25 U/g, 180 min	–	–	–	–	–	–	4.3	[13]
Guava pulp(a)	(f)	25 °C, 30 min	–	–	–	–	–	135	135.6	[14]
GAC fruit peel(d)	(f)	50 °C, 76 min, 250 W	–	–	–	–	–	–	269	[15]
Apricot and shrimp waste(d)	(l)	20 min, 80:20 choline chloride and tartaric acid ratio,	–	76.11	–	–	–	–	–	[16]
Orange peel(d)	(f)	400 W, 30 min, 50% ethanol–water	–	–	–	–	–	–	0.63	[17]
Orange peel(e)	(m)	1:3 solid liquid ratio, 5 min, 200 W	–	–	–	–	–	–	32.08	[34]
Tomato processing waste(e)	(f)	63.4 °C, 20 min, 100 mL/g	–	–	–	1334.8	–	–	–	[18]
Vegetable waste matrix	(h)	59 °C, 30 min, 350 bar, 15 g/min CO_2	–	–	–	–	–	90 (w/w)	–	[19]
Heterochlorella luteoviridis microalgae	(n)	40 °C, 75% ethanol	–	–	–	–	–	86	–	[20]
Xanthophyllomyces dendrorhous yeast(e)	(f)	35 °C, 154 W,	–	–	–	–	–	90	680	[21]
Carrot peels	(h)	58.5 °C, 306 bar, 14.3% ethanol	–	–	–	–	–	86.1	–	[22]
Pumpkin peel(e)	(f)	203 W, 30 min, 31 mL/g solvent to solid ratio	–	–	–	–	237.6	–	–	[23]
Corn gluten meal(e)	(h)	80 °C, 6500 psi, 15 % ethanol	–	–	73.5	–	–	–	–	[24]
Rhodotorula glutinis yeast(e)	(k)	25 °C, 24 h, pH 8	–	–	–	–	–	–	375	[25]
Citrus fruit waste(b)	(f)	208 W/cm^2, 20 °C, 5 min	–	–	–	–	–	–	11.25	[26]
Dunaliella salina algue(e)	(h)	55 °C, 400 bar	–	–	–	–	–	–	115.435	[27]
Chlorella vulgaris microalgae	(o)	110 °C, 30 min	–	–	–	–	–	45	–	[28]
Pomegranate waste (d)	(f)	51.5 °C, 30 min, 0.1 peel to solvent ratio	–	–	–	–	–	–	0.3255	[29]
Citrus byproducts(a)	(h)	44.88 °C, 25.196 MPa	–	–	–	–	–	–	1.91	[53]
Heterochlorella luteoviridis microalgae(a)	(f)	30 °C, 50 W/cm^2	–	–	–	–	–	–	1.31	[30]
Red capsicum(d)	(k)	60 °C, 1 hr	–	–	–	–	–	–	279.83	[31]
Lycium ferocissimum Fruits(e)	(o)	70 min, 70 solvent to solid ratio	–	–	–	–	–	–	589	[32]

Unit: (a) mg/g (b) mg/L (c) mg/kg/U (d) mg/100 g dry weight (e)μg/g Extraction method: (f) Ultrasound assisted extraction (g) Extraction with edible oils (h) Supercritical CO_2 extraction (i) Microwave assisted extraction (j) Monosonication (k) Enzyme-assisted extraction (l) Natural deep eutectic solvents assisted microwave extraction (m) Ionic liquid assisted ultrasonic assisted extraction (n) Moderate electric field (o) Solvent extraction.

Ionic liquid 1-butyl-3-methylimidazolium chloride with ultrasound-assisted extraction of orange peel lead to maximum total carotenoids recovery of 32.08 μg/g. The extraction with ionic liquid was better than conventional solvent, due to good affinity between 1-butyl-3-methylimidazolium chloride and carotenoids as compared to that of acetone. Ionic liquid also contributes to cell disruption which improves the access to the target compounds. Furthermore, using ultrasound during the extraction process increases the mass transfer and the biomass cell disruption, this leads to greater solvent penetration and capillary effects, resulting in high yield and short extraction time. In addition, the use of ionic liquid leads to amplification in the extraction performance. This makes it valuable extraction technique with low environmental impact [34].

Xanthophyllomyces dendrorhous yeast was extracted to give 680 μg/g total carotenoids content [21]. Trans-lutein from pumpkin peel was extracted to be 237.6 μg/g [23], and total carotenoids from citrus fruit waste were extracted by ultrasonic extraction to be 11.25 mg/L [26]. From pomegranate waste, carotenoids were ultrasound-assisted extracted at optimum conditions of 51.5 °C, 30 min, and 0.1 peel to solvent ratio, to give total carotenoid content 0.3255 mg/100 g dry weight [29]. Ivana et al (2021) used olive

oil as a solvent for extraction of carotenoids from orange peel using ultra-sound assisted extraction at 35 min extraction time, 42 °C temperature and 15 mL/g liquid to solid ratio to obtain the 1.85 mg/ 100 g total carotenoids content [37]. On the other hand, same technique was used to extract, 140.70 mg ß-carotene/100 g of dry sample was obtained from mandarin epicarp [38]. Total carotenoid extracted from *Rhodotorula glutinis* yeast by sonication under pressure at 200 kPa and 2 min of extraction time was found to be 25 mg/L [39]. With a 2:1 M ratio of choline chloride and tartaric acid deep eutectic solvents were used to extract β-Carotene of 76.11 mg/100 g of dry weight at 20 min of extraction time from apricot and shrimp waste [16]. A moderate electric field technique was used to recover carotenoid recovery of 86% from *Heterochlorella luteoviridis* microalgae [20]. In comparison, the pulsed electric field gives total carotenoids content of 375 μg/g from *Rhodotorula glutinis* yeast [25]. From brown seaweeds, extraction with vegetable oils was carried out [10].

2.4. Degradation mechanism of carotenoids

The important carotenoids in plants and foods are lycopene, zeaxanthin, lutein, ß-cryptoxanthin, ß-carotene, α-carotene, etc. Several research papers have reported the losses of carotenoids during the storage and processing of plants and foods. Also, the knowledge regarding the mechanism and reactions is still minimal. Some researchers attempted in the area of degradation mechanism of carotenoids in the 1980s. However, such attempts were redirected into other research areas related to the health and benefits of these colorants.

Synthetic carotenoids can be made using Wittig reaction between two phosphonium salt molecule and one dialdehyde molecule. The product is isomerised to make ß-carotene, lycopene or astaxanthin. The degradation mechanism for synthetic carotenoids is described in Fig. 1. Microalgal ß-carotene is natural blend of all trans and 9-cis isomers with numerous health benefits. On the other hand, synthetic ß-carotene is only made up of trans isomer and does not have same health benefits. Similarly, natural microalgal astaxanthin is highly (greater than 95%) esterified. Synthetic astaxanthin is unesterified. As compared to synthetic astaxanthin, the natural astaxanthin was found to be 50 times more effective at singlet oxygen quenching and 20 times more effective at neutralising free radicals [40]. Synthetic carotenoids are prohibited in organic agriculture [41].

Carotenoids are divided into two parts, carotenes and xanthophylls. While xanthophylls are basically oxygenated carotenoids, containing carbonyl, alcohol or any other functional group, the carotenes are hydrocarbon structures. Epoxy carotenoids are group of xanthophylls, but its physical significance is unclear. The carotenoids group which lacks an end group and has short carbon skeleton are called as apocarotenoids. Hydroxy carotenoids contains at least one hydroxyl group and they are also called as carotenols. Examples of hydroxyl carotenoids are lutein(yellow in color), ß-cryptoxanthin(organe in color), zeaxanthin(yellow-organe in color) etc. The carotenoids has large number of conjugated double bonds because of which it has large number of isomers. Most carotenoids which are present naturally are in *trans* form (E) while few are available in *cis* form (z) for example bixin in annatto seeds. The physical properties of Z-isomers are different as Z-isomerisation shortens the value of lamda max resulting in low visibility and color. As compared to E-isomers, the Z-isomers are more soluble in oil and organic solvents. Z-isomers are also less stable and have lower melting point as compared to E-isomers [42]. The low molecular weight carotenoids or volatiles are colorless and completely lack the biological activity. These compound contribute to the flavour of meals and beverages, either positively or negatively [43].

To improve the bioavailability of the carotenoids, the processing temperature is generally above 100 °C, which leads to formation of isomers. Some research has shown that Z-isomers are not only more bioavailable but also high anticancer, and antioxidant activity [44]. However, it mainly depends upon the type of carotenoids. Z-isomers of astaxanthin and lycopene have more bioavailability than its E-isomers [45]. On the hand, E-isomers of lutein and ß-carotene have more bioavaibility than z-isomers [46].

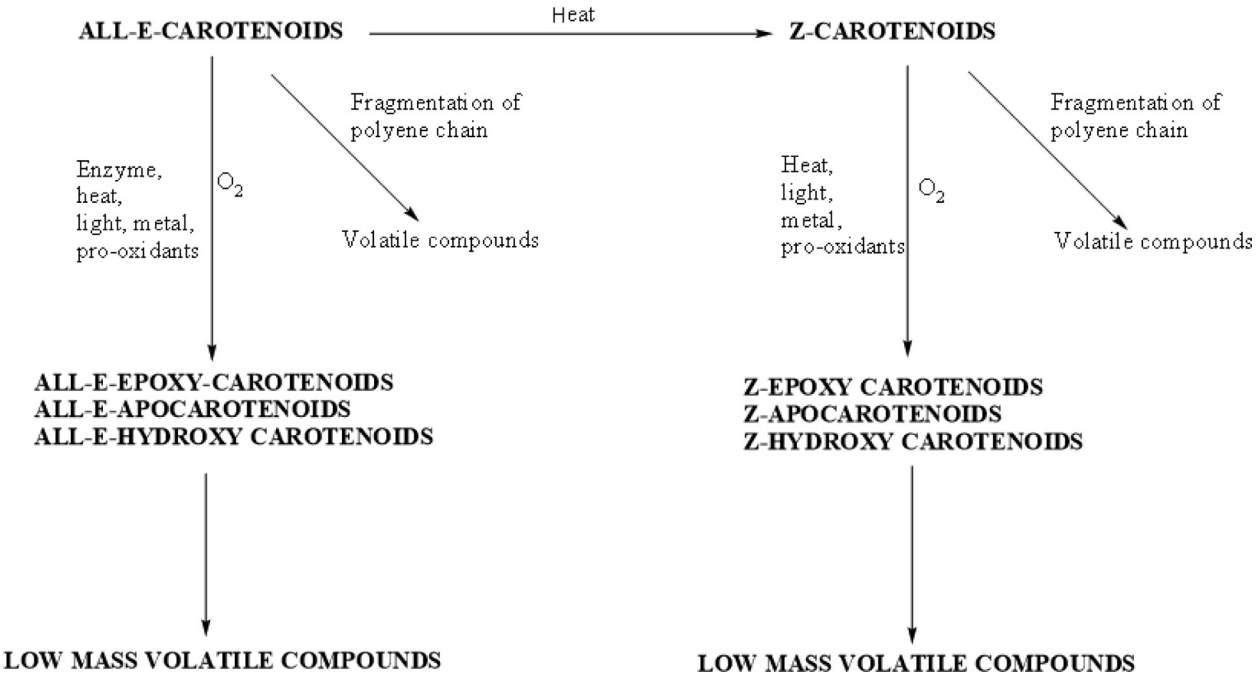


Fig. 1. Degradation mechanism of carotenoids [6].

Recently there has been renewed interest in the degradation mechanism after different findings of positive or negative effects of colorants on the health of the human being. As shown in the Fig. 1, the unsaturated carotenoids majorly undergo oxidation and geometric isomerization. In the initial stage, some parts of E-carotenoids are isomerized to the Z-carotenoids. Both forms of carotenoids are oxidized, starting with epoxidation, apocarotenals, and hydroxylation—more fragmentation results in low mass molecular compounds. Direct conversion by polyene chain into the volatile compounds also takes place [6].

2.5. Recent stability studies of carotenoids

Microencapsulation and nanoencapsulation were investigated to address the problem of stability problems of carotenoids. The recent advances in stability of carotenoids are compiled in Table 2.

As shown in Table 2, encapsulating techniques mostly used are, spray drying, freeze drying, extrusion process, coacervation process. To evaluate the efficiency of the encapsulation technique employed, the encapsulation efficiency is an important parameter. There is an optimum concentration of material to be encapsulated to obtain maximum yield. Also, the retention time and stability of a colorant mainly depends upon the wall material used. Arabic gum is an effective carrier as it has low solution viscosity, high water solubility and good ability to make protective film around the pigment particles. Maltodextrin can also be used as coating material as it has lower cost as compared to Arabic gum and it forms an effective anti-oxygen obstacle. It was found that combination of maltodextrin and Arabic gum was more effective as encapsulating material than their individual use. Spray drying used for encapsulation of mamey and carotenoids in maltrodextrin (10% w/v) and Arabic gum(10%) was found to be more effective as compared to maltodextrin and Arabic gum(5%). The spray dried carotenoids were found to be decrease 30% of reactive oxygen species [47]. Carotenoids from orange peel extracts were encapsulated in alginate beads using extrusion process resulted in encapsulation efficiency of 89.5% with greater antioxidant activity [37]. Oleoresin supercritical extract from sea buckthorn was microencapsulated showed 92% encapsulation efficacy for lycopene and total carotenoids. The cross-linked experiments developed the protein network between casein and protein were able to stabilise the encapsulated compound as high loading capacity was found in whey protein-casein crosslinked network as compared to non-aggreagated network [48]. The microcapsules of red melinjo peels at optimum conditions of 150 °C spray inlet temperature and 100% Arabic gum as coating material with total carotenoids content of 420.40 μg/g and 91.23% of encapsulation efficiency [49]. From tomato peels, lycopene was encapsulated using complex coacervation technique with encapsulation efficiency of 83.6%. For this purpose, researcher's used whey protein isolate and acacia gum material [50]. 82% antioxidant activity and 80% carotenoids retention in the encapsulated powder of Gac peel under optimum conditions of 24.5% total solid containing emulsion, 180 mL/h of feeding rate, and 160 °C using spray drying technique, while mixture of whey protein concentrate and gum Arabic with 7:3(w/w) ratio was used as wall material [51]. Inulin and arabic gum showed excellent potential for tomato pomace extract encapsulation [52]. Gum Arabic and maltodextrin were used in different proportion as wall material for encapsulation of norbixin. Spray drying was used for encaspuation and, maltodextrin: gum Arabic(0:100 proportion) formulation showed the highest encapsulation efficiency of 74.91% [53]. Modified starch and maltodextrin showed a low degradation rate of lycopene during the encapsulation of tomato concentrate [54]. Microencapsulated powder of sea buckthorn extracts using complex coacervation followed by freeze drying resulted in 36.23% encapsulation efficiency. It was found that whey protein isolate by transglutaminase increased the cross linking reaction, which improved the complexes stability [55]. The highest encapsulation capacity of 56% was obtained in the encapsulation of carotenoids from sea buckthorn extracts [56]. Microencapsulated *taxo* proved

Table 2
Recent advances in encapsulation of carotenoids.

Carotenoids' source	Encapsulation technique	Particle size(μm)	Encapsulation efficiency (%)	Special remark	References
Orange peel	Extrusion process	0.78	89.5	Extraction was done using olive oil which resulted in increase in anti-oxidant activity	[37]
Mamey and carrot	Spray drying	–	–	After three months of storage, only 30% reduction in reactive oxygen species	[47]
Sea buckhorn	Complex coacervation and freeze-drying	–	96.40	Crosslinked aggregates showed better release of carotenoids in the simulated intestine study.	[48]
Tomato peels	Complex coacervation and freeze-drying	0.3788–3.56	83.6	The first-order kinetic model fitted for the degradation rate	[50]
Red melinjo peel	Spray drying	2.85	91.23	Best microcapsules were reported at 100% arabic gum material and 150 °C inlet spray temperature	[49]
Gac peel	Spray drying	>10	–	80% retention of carotenoids in encapsulated powder at optimum conditions	[51]
Tomato pomace	Spray drying	0.69–26.51	25.3	Antioxidant activity, particle morphology, and carotenoids loading depend upon wall material used.	[52]
Norbixin	Spray drying	5.018	75	Apart from all blends of wall material, gum arabic gave higher encapsulating efficiency	[53]
Tomato concentrate	Spray drying	–	–	Antiradical activity increased by 2 to 2.4 times in encapsulated product	[54]
Sea buckthorn	Freeze drying	1.26–6.58	36.23	Low degradation rate and good antioxidant activity in stability test	[55]
Sea buckthorn	Coacervation	2–30	56.16		[56]
Banana passion fruit	Spray drying	–	–	Stable composition of phytochemicals and antioxidant activity	[57]
Marigold flower	Spray drying	10–100	95	Improvement in solubility and bioavailability	[58]
Broccoli	Coacervation	65–125	58	The microencapsulation process was affected majorly by encapsulation agent concentration	[59]

to be a suitable food ingredient with excellent bioactive compound [57]. As compared to maltodextrin, capovidone showed better dissolution and antioxidant characteristics [58]. A higher amount of total chlorophyll content was found in the optimized microencapsulation process of lyophilized broccoli [59].

Nano-encapsulation is a promising technique which is capable of improving the water solubility of hydrophobic bioactive components like, carotenoids. It increases the dissolution rate as it works at nano scale(less than 100 nm) particle size. Because of small particle size, water molecules come in contact with larger surface area. Carotenoid-rich extract from Cantaloupe pulp were subjected to encapsulation in porcelain gelatine, whey protein isolate, whey concentrate. Freeze drying was applied and 90% encapsulation efficiency was found for porcelain gelatin was used as wall material with particle size of 59.3–161 nm [60]. Carotenoid-rich extract from cantaloupe melon were nanoencapsulated using porcine gelatine and freeze drying method to obtain 94.8% encapsulation efficiency and 90.9 nm mean particle size. The results indicated that the encapsulating agent was mainly responsible for enhancing and maintaining the antioxidant potential stability of carotenoids under different storage condition [61].Tomato peel extract were subjected to encapsulation using electrospinning technique into composite zein nanofibres. The encapsulation efficiency was more than 90% and mean particle size of 163.7 nm. Moreover, the antioxidant activity was increased by 11-fold by nanoencapsulation with improvement in thermal and storage stability [62].

3. Anthocyanins

3.1. Introduction to anthocyanins

The anthocyanins are red, purple, or blue pigments found in fruits, flowers, plants, etc. The molecular structure of anthocyanins is based on glucoside with sugar moieties. There are mainly six derived anthocyanins which are cyaniding (50%), pelargonidin (12%), peonidin (12%), delphinidin (12%), petunidin (7%), and malvidin (7%). These anthocyanins are among the 600 types of molecules which can be found naturally [63].

The production of synthetic anthocyanin is possible, but their yield by chemical synthesis is very low. It produces food colorant with consistence purity and known properties. But, synthetic anthocyanins requires extensive toxicological test before getting clearance from food authority. Synthesis of flavylium cation occurs under harsh conditions such as condensation of 2,4-dihydroxy benzaldehyde with different acetophenone derivative with acetic and sulphuric acid. However, due to health concerns related to use of synthetic anthocyanins, the natural anthocyanins are preferred [64].

3.2. Sources of anthocyanins

The current research focus is more oriented towards finding new food, plant sources, and novel extraction techniques by which higher values of anthocyanins can be achieved. New sources from which anthocyanins can be extracted efficiently are red cabbage [63], *hibiscus sabdariffaI* [65], campbell early grape [66], *sorbus pohuashanensis* fruits [67], purple passion fruit [68], mulberry wine residues [69], mango peel [70], rice bran [71], grape pomace [72], wine lees [73], fresh mulberry[74], black soybeans [75], saffron petal [76], rice berry bran [77], haskap berry pulp[78], beibinghong *vitis amurensis rupr* [79], *arbutus unedo L.* fruit[80], cherry [81], *ficus carica L.* peel [82], black carrot pomace [83], purple sweet potato [84], *nephelium lappaceum L.* fruit peel [85], blueberry fruit [86], eggplant peel [87].

3.3. Extraction methods of anthocyanins

As shown in Table 3, novel extraction techniques are under development like DES-assisted microwave extraction, acidified water extraction, enzyme hydrolysis, ultrasound-assisted extraction, ultrasound-assisted enzymatic extraction, enzymatic assisted extraction, natural deep eutectic solvents, supercritical fluid extraction, pulsed electric field extraction, heat-assisted extraction.

Hibiscus sabdariffaI was subjected to DES-assisted microwave extraction at 550 W and 35 mL of citric acid/ethylene glycol with a 1:4 M ratio to obtain the highest extraction of total anthocyanins of 2.961 mg cyaniding-3-glucoside equivalent /g of dried sample [65]. The highest anthocyanin acquisition rate of 84.82% from the blueberry powder was obtained at 1st stage microwave power of 800 W and 2nd stage microwave power of 280 W [88]. The microstructural analysis observed that microwave irradiation disrupted the cell walls of the saffron flower tepal. 101 mg anthocyanins/g dried sample was extracted at optimum conditions of 48 °C, 9.3 min, and 77.5 mL/g [76]. Anthocyanin extraction reached 1.24 folds by natural deep eutectic solvents (6.05 mg/g fresh weight) than organic solvent from mulberry [74]. *Dacryodes rostrata* peel was subjected to ultrasound-assisted extraction at 30 °C, 10 min, and pH 6.0 to give total anthocyanins of 494.8 mg/g [89]. Mulberry wine residues were extracted using ultrasonic-assisted enzymatic extraction to give 5.98 mg/g of total anthacyanins [69]; in comparison, 306.94 mg total anthocyanins/g rice brain were extracted from rice bran [71]. At optimum conditions of 81.4% amplitude, the solid–liquid ratio of 49.1 g/mL, 8.59 min, the ultrasound-assisted extraction of black soybeans, the total anthocyanins content was 66.44 mg/100 g [75]. As compared to solvent(ethanol and D-limonene) extraction, ultrasonic-assisted extraction of Thai rice berry bran gave higher values of total anthocyanin content(42.50 μg/g), total flavonoid content(27.89 mg/g), and total phenolic content(109.71 mg/g) [77]. *Ficus carica* L. peel was extracted by three methods: heat, microwave, and ultrasound-assisted extraction. Out of which, ultrasound-assisted extraction provided the highest anthocyanin content (3.81 mg/g) comparatively [82]. Ultrasound-assisted extraction of purple and black rice bran gave higher values of total anthocyanin content, total phenolic content than conventional extraction process [90]. With optimum conditions of 50 °C and 183.1 J/g energy density, ultrasound-assisted extraction of black pomace was found to obtain higher values of cyanidin-3-xyloside-galactoside-glucoside-ferulic acid and cyanidin-3-xyloside-galactoside [83]. In contrast, Ultrasound-assisted ammonium sulfate/ethanol aqueous two-phase system was used for purple sweet potato extraction at 52.48 °C and 48.19 min of optimum conditions to get 191.09 mg/L of total anthocyanin. The natural colorant extracted was applied to dye silk fabric. This method was found to avoid severe environmental pollution problems and decrease the use of solvent [84]. Higher values of total anthocyanin (10.26 mg/100 g) and phenolics (552.64 mg GAE/100 g) were obtained from *Nephelium lappaceum* L. fruit peel [85]. Eggplant peels were subjected to ultrasound-assisted extraction to give the highest anthocyanins content (2410.71 mg/kg) at 55.1 °C, 44.85 min, 37 kHz, and 54.4% methanol concentration [87].

From haskap berry pulp, 52.7% anthocyanin extraction yield was obtained by supercritical CO_2 extraction with optimum conditions of 65 °C, 15 min static and 20 min dynamic time, and 45 MPa [78]. The pulsed electric field can efficiently and rapidly extract several types of anthocyanins from *V. amurensis Rupr* [79]. The pulsed electric field was found to positively affect blueberry fruit juice yield and anthocyanin content [86]. At the optimum condition of 80° C, 10 min, 1/50 g/mL, Campbell Early grape were extracted employing acidified water extraction to obtain total anthocyanins of 197.52 mg/100 g, total phenols of

Table 3
Recent advancement in the extraction of anthocyanin.

Source (Unit of concentration)	Extraction method	Extraction conditions	Total Anthocyanins	Anthocyanin yield (%)	References
Jambolan fruit(a)	Ultrasound assisted extraction	5000 W/L, 7.5 min, 79.6 % ethanol	60.5	–	[95]
Blueberry and chokeberry(a)	Subcritical water extraction	190 °C and 1 min(for chokeberry) 130 °C and 3 min(for blueberry)	0.66 0.47	–	[96]
Hibiscus sabdariffal(a)	DES assisted microwave extraction	550 W power, 35 mL DES including 50% water (v/v)	2.961	–	[65]
Campbell Early grape (b)	Acidified water extraction	80 °C, 10 min, 1/50 g/mL	197.52	–	[66]
Red cabbage(c)	Extraction and purification	24 °C, 12 h	3221.45	–	[63]
Sorbus Pohuashanensis Fruits	Enzyme hydrolysis	50° C , 2.5 h, pH 3.0, pectinase 0.15%	–	15.33	[67]
Dacryodes rostrata peel (a)	Ultrasound assisted extraction	30° C , 10 min, pH 6.0	494.8	–	[89]
Purple passion fruit (b)	Conventional method	52.03 °C,180 min, pH 2.0, 29.79 g of solvent per g of raw material	577.59	–	[68]
Mulberry wine residues(a)	Ultrasonic-assisted enzymatic extraction	52 °C, 94 min, 315 W, 0.22% enzyme	5.98	–	[69]
Mango peel(b)	Sequential extraction	67.45% ethanol,60 min, 0.24% acetic acid	22.33	–	[70]
Rice bran(a)	Ultrasonic-assisted extraction	60 °C, 60 min	306.94	–	[71]
Blueberry powder	Variable power microwave extraction	800 W	–	17.75	[88]
Grape pomace (d)	Enzymatic Assisted Extraction	40 °C, 0.25% enzyme preparation percentage	2.67	–	[72]
Wine lees(a)	Conventional method	25 °C, 15 min, 50% ethanol	6.20	–	[73]
Fresh mulberry(a)	Natural deep eutectic solvents	liquid–solid ratio 22 mL/g, 30 min, mole ratio of 1:1:1, 30% water content	6.05	–	[74]
Black Soybeans(b)	Ultrasound-Assisted Extraction	81.4% amplitude, solid–liquid ratio of 49.1 g/mL, 8.59 min	66.44	–	[75]
Saffron tepal(a)	Microwave-assisted extraction	48 °C, 9.3 min, 77.5 mL/g	101	–	[76]
Riceberry bran	Ultrasonic assisted extraction	51 °C, 40 min	42.50	–	[77]
Haskap berry pulp	Supercritical fluid extraction	65 °C, 15 min static and 20 min dynamic time, 45 MPa	–	52.7	[78]
Beibinghong *Vitis amurensis Rupr* (b)	Pulsed electric field extraction	15 kV/cm, 8.5 w/w solid to liquid ratio	166.65	–	[79]
Arbutus unedo L. fruit (e)	Heat assisted extraction	90 °C, 5 min, 80% of ethanol	–	51.2	[80]
Edible portion of cherry(b)	Solvent extraction	37 °C, 90 min, 10 mL/g solvent to solid ratio	244	–	[81]
Ficus carica L. peel(a)	Ultrasound extraction	310 W, 21 min, 100% of ethanol	3.82	–	[82]
Purple rice bran(f)	Ultrasound extraction	31.7 °C, 15.77 min, 2.39 pH	34.86	–	[90]
Black rice bran(f)		35.97 °C, 22.89 min, 2.52 pH	31.95	–	
Black carrot pomace	Ultrasound extraction	50 °C, 183.1 J/g energy density	–	40.6	[83]
Purple sweet potato(f)	Ultrasound-assisted ammonium sulfate/ethanol aqueous two-phase system	52.48 °C, 48.19 min	191.09	–	[84]
Nephelium lappaceum L. fruit peel(b)	Ultrasound extraction	50 °C, 20 min, 20 W, 1:18.6 g/mL	10.26	–	[85]
Blueberry fruit(f)	Pulsed electric field	3 kV/cm, 24 h	215.8	–	[86]
Black carrot juice(c)	Enzyme assisted extraction	60 °C, 78 min, 0.23% (v/w) enzyme concentration	1252.15	–	[94]
Saffron petals(f)	Acidified ethanol solvent extraction	25.8 °C, 24 h, 20 mL/g	1609.11	–	[91]
Black carrot(a)	Acidified ethanol solvent extraction	50 °C, pH 3.5, ethanol/water ratio 75:25 (v/v), solvent solid ratio 10:1 (v/w)	1.1	–	[92]
Ixora siamensis(f)	Acidified ethanol solvent extraction	80 °C, 143.21 min, 0.5% Trifluoroacetic acid content	5670.67	–	[93]
Eggplant peel	Ultrasound-assisted extraction	55.1 °C, 44.85 min, 37 kHz, 54.4% methanol concentration	2410.71	–	[87]

(a) mg cyanidin-3-glucoside equivalent /g of dried sample (b) mg/100 g (c) mg cyanidin-3-glucoside equivalent /mL (d) g/100 g (e) μg/g (f)mg/L (g)mg/kg.

1363.12 mg/100 g, and 2,2′-azino-bis (3-ethylbenzothiazoline-6-sulfonic acid) (ABTS) radical scavenging activity of 315.79 mg/100 g [66]. Maximum monomeric anthocyanin content (1609.11 mg/L) was obtained at 25.8 °C, 24 h, 20 mL/g from saffron petals, which were found to be enriched in anthocyanins and could be used in pharmaceuticals and food industries [91]. Optimum black carrot extraction conditions were 50 °C, pH 3.5, ethanol/water ratio 75:25 (v/v), and solvent solid ratio 10:1 (v/w), which lead to lower polymeric color, higher anthocyanin content, antioxidant activity, and phenolic content [92]. The acidified ethanol solvent extraction of *Ixora siamensis* at optimal combination at 80 °C, 143.21 min, 0.5% Trifluoroacetic acid content resulted in higher anthocyanin(5670.67 mg/L) content [93]. The red cabbage was first extracted by acidified water at 24 °C, and 12 h of extraction time, and then it was subjected to purification step of micro, ultrafiltration, and adsorption processes. Because of this, the pigment concentration was three times more than that of the initial concentration [63]. Ethanol-based extraction was carried out of mango peel to obtain 22.33 mg/100 g of total anthocyanins and 132.63 mg/g of total phenolic content [70]. *Sorbus Pohuashanensis* Fruits were subjected to enzyme hydrolysis at 50 °C, 2.5 h, pH 3.0, and pectinase 0.15% to get an extraction yield of 15.33%. Also, the extract was purified by D101 macroporous resin, which showed a 93.75% of adsorption rate [67]. At lower temperatures, 40 °C, and 0.25% enzyme preparation percentage encouraged anthocyanin extraction of 2.67 g/100 g from grape pomace [72]. Enhanced values of anthocyanins (1252.15 mg/L) from black carrots by enzyme-assisted extraction at 60 °C, 78 min, and 0.23% (v/w) enzyme concentration. Overall twofold anthocyanin recovery was obtained as compared to untreated mash [94].

577.59 mg/100 g total anthocyanins were extracted from purple passion fruit by conventional extraction method at 52.03 °C, 180 min, pH 2.0, 29.79 g of solvent per g of raw material [68]. When microwave was used as pretreatment for conventional treatment of wine lee at 25 °C, and 15 min doubled the total anthocyanin content(6.20 mg/g) compared to ultrasonic as pretreatment [73]. 244 mg/100 g total anthocyanins were determined from fresh cherry by ultra-performance liquid chromatography-mass spectroscopy analysis [81]. At 90 °C, 5 min, 80% of ethanol, 51.2% extraction yield was obtained from *Arbutus unedo L.* fruit by heat-assisted extraction [80]. 244 mg/100 g total anthocyanins were determined from fresh cherry by ultra-performance liquid chromatography-mass spectroscopy analysis [81]. Probe ultrasound method resulted in highest recovery of total anthocyanins as compared to pressurized liquid extraction as there was more cell breakdown [95].

3.4. Degradation mechanism of anthocyanins

While the color of anthocyanins depends upon the pH, the stability depends on other factors like temperature, oxygen, structure and concentration, light, pH, solvents, protein, presence of enzymes, metallic ions [6]. As shown in the Fig. 2, anthocyanins can exist in different structural configurations depending upon the pH. At pH 1–2, flavylium cation is the dominant species. Furthermore, the quinoidal base prevails at pH 2–4. Colorless carbinol pseudo base is the dominant species at pH 4–5, and pale yellow chalcone prevails at pH 6 [6].

3.5. Recent stability studies of anthocyanins

The stability of anthocyanins appears to be a significant problem, and for that purpose, various microencapsulation studies have been conducted as shown in Table 4. From the grape pomace extract, encapsulation by spray drying and freeze-drying method presented the highest values of phenolic compound (2.66 mg/g and 2.42 mg/g) retention and total anthocyanins(0.346 and 0.353 mg/g malvidin-3,5-diglycoside equivalent). Also, simulated gastrointestinal digestion assay showed that powder that contained whey protein isolate and gum arabic as wall materials had more release of anthocyanins in the intestinal phase [97]. Microencapsulation proved to be effective in encapsulating anthocyanins from purple flesh cultivated potato against environmental conditions. The microparticles of purple flesh potato showed 80% encapsulation efficiency under optimized conditions [98]. The spray-dried microcapsules of grape skin demonstrated low degradation constant (0.0207 h^{-1}) and a high half-life (33.47 h) during treatment in the dark. Also, microencapsulating with emulsification with drying was more feasible for increasing the stability of anthocyanins [99]. From eggplant peel, microencapsulated bioactive powder by freeze-drying method showed the 94.31% retention of anthocyanins in extract [100]. Pure whey protein spray-dried microencapsulated from black rice released the highest values of total phenolic content and antioxidant activity in the simulated gastrointestinal fluid study [101]. Red flesh apple extract was encapsulated with co-pigmentation and spray drying methods resulting in improved anthocyanins stability [102]. In vitro study found that blueberry anthocyanins retained inside carboxymethyl starch and xanthan gum in the stomach and liberated in the intestine [103]. The atomization technique gave better efficiency and antioxidant capacity of anthocyanin hibiscus extract as it produced small particles [104]. The optimal microcapsules of black raspberry extracts showed intense red color throughout storage, showing the efficacy of the complex coacervation process of encapsulation [105]. Supercritical CO_2 showed promising results for the encapsulation of amphiphilic and hydrophilic compounds in liposomes [106]. Cress seed gum was found to have excellent encapsulation efficiency when microencapsulation of saffron petals extract was done by freeze-drying method [107]. Spray drying of barberry extract with maltodextrin and gum Arabic as wall material led to higher encapsulation efficiency [108]. Maltodextrin and arabic gum at 20 to 80 ratio gives highest encapsulation of anthocyanin rich red cabbage extract with good anthocyanin content and solubility [109].

Nano encapsulation was employed to sour tea extract by maltodextrin and whey protein concentrate to obtain encapsulating efficiency of 67.3% and particle size of 139.03 nm. The encapsulation increased the antioxidant properties, reduced oxidative spoilage process in chicken nugget and increased shelf life of chicken nugget [110]. Anthocyanin was encapsulated in peptide nanocomposite with 77.06% encapsulation efficiency and less than 100 nm particle size. It was found that co-assembly of peptide with anthocyanin causes conformational change from α-helical structure to ß-sheet, which makes encapsulation of anthocyanin possible. The C6M1 peptide nanocomposite improves the tolerance against high temperature, pH and metal ions [111]. By pectin and lysozyme self-assembling, the blackberry extract were nanoencapsulated with 73% efficiency and 200–300 nm particle size [112].

4. Betanins

4.1. Introduction to betanins

With nitrogen in their basic structure, betanins are water-soluble nature pigment. Betanins are divided into two structural groups: betacyanins (responsible for red-violet color) and betaxanthins(responsible for yellow-orange color) [114].

4.2. Sources of betanins

The different investigator found novel sources from which betanins can be extracted like, *Gomphrena globosa* L. flower [115], yel-

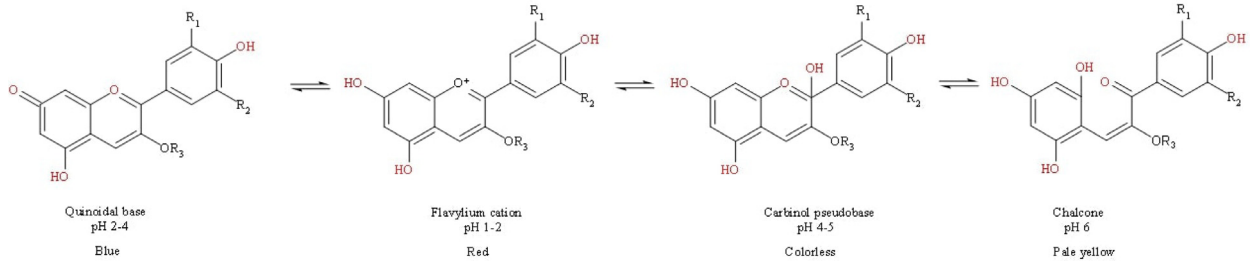


Fig. 2. Degradation mechanism of anthocyanins [6].

Table 4
Recent development of microencapsulation techniques for anthocyanins.

Anthocyanin source	Encapsulation technique	Particle size(μm)	Encapsulation efficiency (%)	Special remark	References
Red cabbage	Drum drying	–	98.85	Drum drying technique prevents food anthocyanins to degrade.	[109]
Hibiscus sabdarifa	Spray drying and freeze drying	–	–	Powders can be used as bioactive ingredient because of its higher solubility and thermal stability	[113]
Grape pomace	Spray drying and freeze-drying	–	–	The sample that did not contain Pec showed a higher release of anthocyanins	[97]
Purple flesh cultivated potato	Spray drying	1.95–20.9	86	Maltodextrin as an encapsulation agent increased bioaccessibility by 20%.	[98]
Grape skin	Internal gelation with Spray drying and freeze-drying	0.01–3000	75	The degradation of anthocyanins followed first-order kinetics	[99]
Eggplant peel	Freeze drying	1–3	94.31	The addition of microencapsulated bioactive compounds increased chewiness and decreased the firmness of the sample.	[100]
Black rice	Spray drying	1–30	100	88% of the highest anthocyanin retention by maltodextrin microcapsule.	[101]
Red-flashed apple	Copigmentation and spray drying	0.32–0.34	96.85	No significant difference between spray and freeze-drying methods on stability improvement.	[102]
Blueberry	Freeze drying	–	97.16	Carboxymethyl starch and xanthan gum combination increases the stability and target release in the intestine	[103]
Hibiscus sabdariffa L.	Ionic gelation	78–1100	93.9	The Arrhenius equation was followed by anthocyanin degradation. Also, better stability with temperature variation.	[104]
Black raspberry	Double emulsion technique	35.34–80.22	–	Better thermal and storage stability	[105]
Soy lecithin	Supercritical CO_2 method	0.159	78.2	Anthocyanins were released slowly in the simulated gastric fluid, while rapid release and degradation were evident in simulated intestinal fluid.	[106]
Saffron petals	Freeze drying	–	–	Cress seed gum was found to have a better protective effect	[107]
Barberry	Spray drying	–	96.215	Maltodextrin and gum Arabic showed high efficiency	[108].

low pitaya [116], dragon fruit peel [117], red beetroot [118], beet leaves [119], *Alternanthera sesillis* [120], beetroot pomace [121], *Chenopodium quinoa wild* hulls [122], *Alternanthera brasiliana* [123], floral bracts of *Bougainvillea glabra* [124], beta Vulgaris peels [125], prickly peer fruits [126], waste red beet stalks [127], prickly peer peels [128].

4.3. Extraction methods of betanins

Apart from conventional extraction, various novel extraction techniques like ultrasound-assisted extraction [115], aqueous two-phase system [116], adsorption [118], microwave-assisted extraction [125], pulsed electric field [128] were investigated for the extraction of betanins from novel sources.

Betacyanin was extracted from *Hylocereus polyrhizus* using pectinase assisted extraction with ethanol as a solvent and the extract was used as natural colorant in ice cream [129]. *Gomphrena globosa* L. flower was subjected to ultrasound-assisted extraction to obtain 161.6 mg/g of betacyanin content. The betacyanin extract was found to be a better alternative as a pink color for cookies than synthetic colorants [115]. At optimum conditions(60 °C, 20 min, 25:1 mL/g, 60% solvent concentration), ultrasound-assisted extraction was found to be excellent at extracting total betacyanin from dragon fruit peel with the highest extraction yield and minimum degradation of the pigment [117]. β-cyclodextrin-enhanced ultrasound-assisted extraction of red beet showed that β-cyclodextrin increased the stability and the highest values of betacyanin (2.243 mg/g) and phenolic compounds (20.03 mg/g) [130]. Betacyanin (71.95 mg/g) and betaxanthin (42.3 mg/g) was extracted using microwave assisted extraction from *Amaranthus tricolour* leaves at 450 W, 60 °C, 15 min extraction time[131]. While the aqueous two-phase system was used to get the maximum recovery of betaxanthin (52.3%) [116]. By demineralization, deproteinization, and deacetylation processes of the shrimp waste, chitosan was extracted and used for adsorption of betacyanin and betaxanthin from red beetroot [118]. While extraction of beet leaves was optimized (61 °C, 80% ethanol, 1:10 mg/mL) by ethanol extraction to obtain higher values of betanin (0.78 mg/g) and betaxanthin (2.17 mg/g) [119]. Ethyl acetate leaf extract of *Alternanthera sesilli* was subjected to solid–liquid extraction to find betalin of 67.08 mg/g, betaxanthin of 53.92 mg/g, and amaranthine of 93.94 mg/g [120]. From beetroot pomace, the conventional extraction was optimized (50 °C, 10 min, 2.5 pH) to get the highest values of betacyanin (17.07 mg/L) and betaxanthin (15.04 mg/L) [121]. Ginger candy rich in beetroot pomace extract was developed under optimum conditions of 7.81 blanching time and 9.24 %beetroot pomace extract with the highest betacyanin (18.79 mg/kg) and betaxanthin (12.75 mg/kg) content [132]. The stems and leaves of *A. brasiliana* and *A. tenella* can be used as an excellent agro-industrial product as well as natural dye additive with decent values of betacyanins content(89.40 μg/g) when extracted with 1 mL of acetonitrile: water acidified with 0.1% formic acid stirred for 60 s [123]. Extraction of water-soluble red pigment from floral bracts of *Bougainvillea glabra* at optimized conditions of 22.5 °C, 3 g, and six h [124]. Natural pigment extraction from red beet waste stalk was subjected to ultrasound-assisted extraction to get the highest yield of betacyanin (1.28 mg/g) and betaxanthin (5.31 mg/g) content [127]. For citric acid, the microwave-assisted extraction of beta Vulgaris peels at optimum condition (224.61 MW, 5.20 pH, 57.06 s) resulted in a high betanin concentration (472.113 mg/L) [125]. The pulsed electric field was better than ultrasound-assisted extraction of prickly pear peels and pulp [128] (see Table 5).

Table 5
Recent advancement in the extraction of betanins.

Source (Unit of concentration)	Method of extraction	Extraction condition	Total betacyanin content	Total betaxanthin content	Total betaxanthin content(%)	Total indicaxanthin content	Reference
Hylocereus polyrhizus(f)	Pectinase-assisted extraction	1.5% pectinase concentration, 95% ethanol	130.83	–	–	–	[129]
Amaranthus tricolour leaves	Microwave assisted extraction	450 W, 90 °C, 15 min	71.95	42.30	–	–	[131]
Gomphrena globosa L. flower(a)	Ultrasound assisted extraction	500 W, 22 min	161.6	–	–	–	[115]
Yellow pitaya	Aqueous two phase system	25 °C, PEG 1000-Phosphate system	–	–	52.3	–	[116]
Dragon fruit peel(a)	Ultrasound-assisted extraction	60 °C, 20 min, 25:1 mL/g, 60% solvent concentration	1.586	–	–	–	[117]
Red beetroot(a)	Adsorption by chitosan extracted from shrimp waste	5.12 pH, 0.02 adsorbent amount	144.15	114.50	–	–	[118]
Red beetroot(a)	Ultrasound-assisted extraction	80 W, 30 min	2.243	–	–	–	[130]
Beet leaves(a)	Solvent extraction	61 °C, 80% ethanol, 1:10 mg/mL	0.78	2.17	–	–	[119]
Alternanthera sesillis(a)	Solid-liquid extraction	30 °C, 24 h, 1:20 mg/mL	67.08	53.92	–	–	[120]
Beetroot pomace(b)	Conventional extraction	50 °C, 10 min, 2.5 pH	17.07	15.04	–	–	[121]
Beetroot pomace(c)	Conventional extraction	50 °C, 10 min, 2.5 pH, 1:15 mg/mL	18.79	12.75	–	–	[132]
Red beet(a)	Ultrasound assisted extraction	52 °C, 90 min, 25% ethanol	4.2	2.8	–	–	[133]
Chenopodium quinoa willd hulls(d)	Ultrasound assisted extraction	91% amplitude, 0.7 cycle, 40 s short extraction time	201.01	–	–	–	[122]
Alternanthera brasiliana	Conventional extraction	0.1% formic acid(1:1,v/v) in acetonitril:water mixture, 60 s	89	–	–	–	[123]
Beta vulgaris peels(b)	Microwave assisted extraction	224.61 MW, 5.20 pH, 57.06 s	472.113	–	–	–	[125]
Prickly pear fruits(c)	Conventional extraction	28 °C , 30 min	–	45.87	–	56.53	[126]
Waste red beet stalk(a)	Ultrasound assisted extraction	89 W, 53 °C, 35 min, 1:19 g/mL	1.28	5.31	–	–	[127]
Prickly peer peels(d)	Pulsed electric field	20 min, 7 pH,	75	–	–	–	[128]

Unit: (a) mg/g of extract (b) mg/L (c) mg/kg (d) mg/100 g (e) μg/g (f) g/kg.

4.4. Degradation mechanism of betanins

As shown in shown Fig. 3, the degradation path of betanin along with the color of the degradation product. These degradation pathways can be divided into five categories: (1) Deglycosylation takes place under the action of high temperature, low pH, or ß-glucosidase, wherein the sugar moiety is eliminated from betacyanin glycoside. Due to respective aglycone release, deep color forms, and the bathochromic shift takes place. (2) Isomerization takes place under the action of thermal procedure, acidic, or alkaline conditions. It is due to stereo-chemistry conversion in which the number of atoms remains the same but the arrangement of atoms changes. Isomerization of betanin at C15 produces isobetanin. Now, isobetanin and betanin both have almost indistinguishable chromatic properties and therefore making them difficult to detect. (3) At a pH greater than six and heating process, the hydrolysis of betalains occurs by aldimine bond cleavage to get the degradation products betalamic acid (yellow color) and cycloDOPA-5-O-ß-glucoside (colorless). As aldimine bond cleavage is reversible at pH 6, these intermediates can recondense to the original molecule. This can be the reason why betanin regeneration takes place after poor heating treatment at low temperatures. (4) Under anaerobic conditions and heating, the dehydrogenation of betacyanins takes place to give neobetanins (yellow color). Here, the double bond formation takes place at betalamic acid moiety. (5) In decarboxylation, the carboxyl groups at C2, C15, and C17 in betacyanins and at C11 and C13 in betaxanthins are eliminated. It is considered beneficial if decarboxylation occurs at C2 or C15, as the chromatic characteristics of betanin are kept. On the other hand, when it occurs at C17, these characteristics vary, and red-violet coloration exhibits [114].

4.5. Recent stability studies of betanins

Because of the easy degradation of betanins, a recent study has been majorly focused on finding different ways to increase its stability to enhance its commercial applications. The use of different additives like chelating agents and antioxidants has been reported in the literature. However, encapsulation has emerged as an excellent method for stability for betalains. Recent work carried out for stability of betacyanins is compiled in the Table 6.

The maltodextrin and chia mucilage had good retention of betacyanin, excellent pH, and thermal stability [134]. Pumpkin protein isolate showed 92% encapsulation efficiency for freeze-drying [135]. The betalain-rich extract of *Opuntia ficus indica* fruit with ionic gelation with calcium alginate showed good stability of color in gummy candies [136]. Spray drying microencapsulation of cactus pear fruit results in a reduction of extract volume, making it easy to deliver and storage [137]. The potato succinylated starch-based microencapsulation of pitaya extract showed good stability for 32 days at 4 °C and 4.6 pH. Good retention of betacyanin content (15%) and total phenolics (60%) was encapsulated in Ca(II)-alginate beads from the beetroot extract [138]. At optimum conditions of 10% w/w maltodextrin, 165 °C drying temperature, and 47 mm air flow rate, the 100% encapsulation efficiency and 58.1% encapsulation yield were found for microencapsulation of organic quinoa [139]. Maltodextrin encapsulation of pitaya fruit extract showed an excellent half-life after storage [140]. Spray drying of

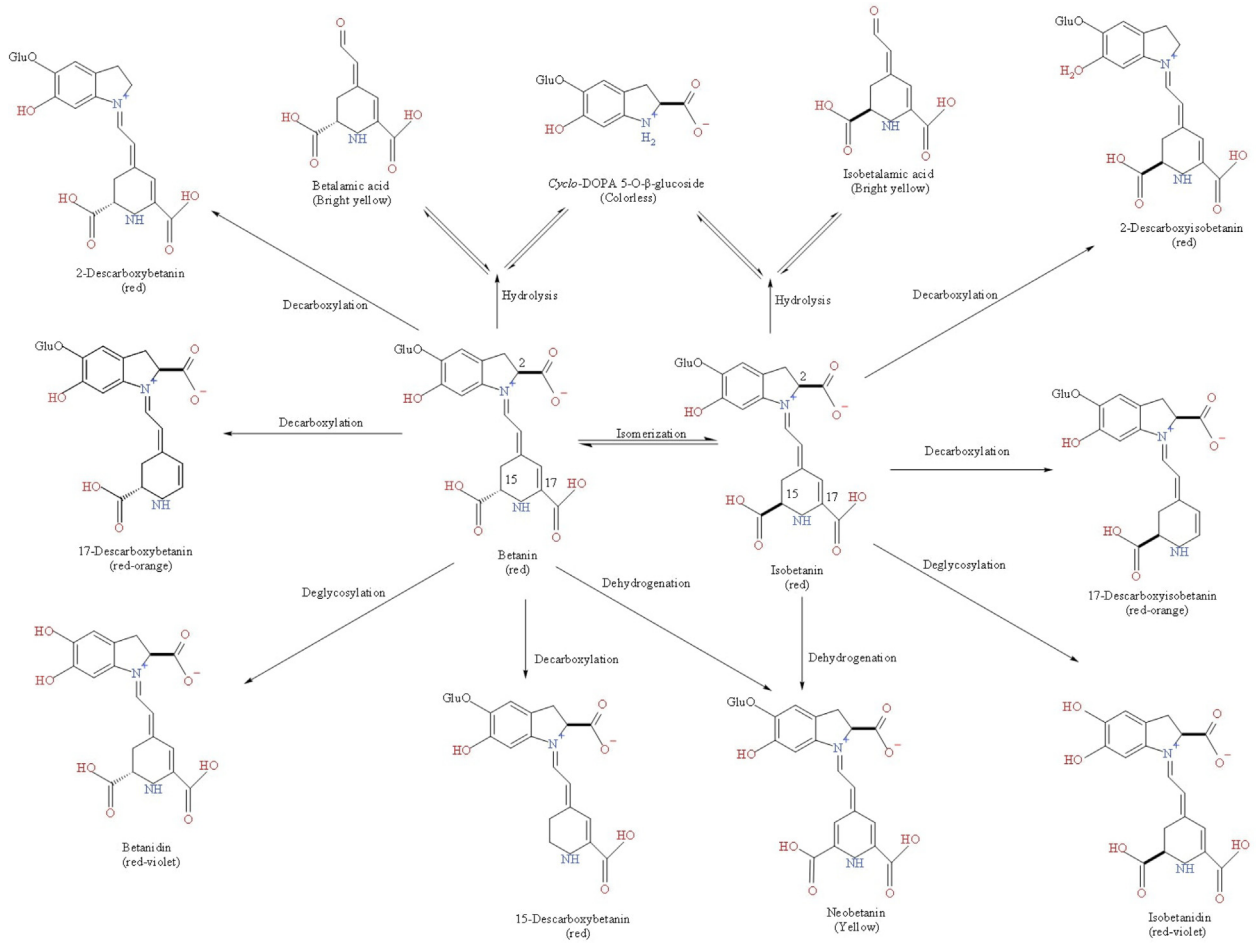


Fig. 3. Degradation mechanism of betanin [114].

Table 6
Recent advances in the stability of betanin.

Betanin source	Encapsulation technique	Particle size(μm)	Encapsulation efficiency (%)	Special remark	References
Red dragon fruit peel	Ionic gelation with sodium alginate	106.32	78.62	For delivery of microcapsules, alginate loaded extract showed good behavior.	[144]
Chia seed	Freeze drying	–	–	Chia mucilage is an excellent encapsulating agent	[134]
Beetroot juice	Freeze and spray drying	4.48–201.37	91.80	Pumpkin oil cake protein as a new encapsulating agent	[135]
Opuntia ficus indica fruit	Ionic gelation	–	–	No variation in color parameter after 30 days of storage	[136]
Escontria chiotilla and *Stenocereus queretaroensis* pulp and skin	Spray drying	–	–	After three months of storage, *Opuntia ficus indica* mucilage was found to be better wall material	[147]
Cactus pear fruit	Spray drying	–	71	Betaxanthin degradation in the presence of light	[137]
Pitaya fruit	Spray drying	4.66–6.93	97.56	Storage stability for 32 days in yogurt at 4.6 pH	[148]
Bougainvillea spectabilis bracts	Gelation	–	88.63–89.82	The extraction solvent did not affect extract stability	[149]
Beetroot leaves and stems	Ionotropic gelation	–	16–60	Simple, cost-efficient method	[138]
Chenopodium quinoa Willd.	Spray drying	3–10	100	Low saponin concentration and high betacyanin content in microcapsules	[139]
Pitaya fruit	Spray drying	–	–	Maltodextrin/pectin resulted in wellformed and homogenous particles	[140]
Pitaya fruit	Spray drying	–	–	Higher storage stability in maltodextrin encapsulated betanin	[141]
Beetroot pomace	Freeze drying	–	86.14	Soy protein isolate as a cheap and readily available source of wall material	[142]
Red dragon fruit	Freeze drying	–	90–95	Carbohydrate encapsulation improved antioxidant and anti-inflammatory	[143]

pitaya powder at a low drying temperature of 150 °C showed a better yield of 59% and excellent red-violet color retention [141]. Beetroot pomace extract encapsulation by the freeze-drying method in soy protein showed higher values of retention of polyphenol (76.67%), betacyanin (17.77%), and betaxanthin (17.72%). Also, the release of phenolics was better in the simulated intestinal fluid than in gastric fluid [142]. Carbohydrate encapsulation of red dragon fruit extract can better stabilize at various pH levels, water

activity levels, and at higher temperature. Also, it carbohydrate encapsulation showed to enhance the antioxidant, and anti-inflammatory effect by two to five fold [143]. From *Hylocereus polyrhizus*, pectinase at a concentration of 1.5% gave the highest betacyanin concentration of 130.83 g kg^{-1}, and low sugar content of 59.74 g/kg was obtained. The application of the betacyanin in the ice cream was found to be better than the commercial colorant (E-162) throughout 21 days of frozen storage [129]. Also, from red dragon fruit, the encapsulation in alginate microbeads was efficiently done with an encapsulation efficiency of 78.62%, a mean particle size of 106.37 μm. The release behavior of encapsulated extract was two folds times the non-encapsulated extract [144]. Stability study of betacyanin pigment from *Basella rubra* showed that lower temperature of 4 °C, absence of light and oxygen seems to improve the stability [145]. Alginate loaded red dragon fruit extract were microencapsulated with a ratio of 1:8.97 alginate: beacyanins showed good antioxidant activity as compared to non-alginated extract [144].

Nanocapsulation in soybean lecithin of red beetroot, opuntia, and red radish was carried out to obtain encapsulation efficiency of 98.8%, 90.4% and 99.9% respectively. It was found that non-encapsulated extract of red radish were effective for coloring yogurt-like fermented soy. While encapsulated opuntia extract can be used as supplement to yogurt like fermented soy with health benefit and without change in color [146].

5. Conclusions

The food colorants industry is growing at a much rapid annual rate of 4.6% and is estimated to achieve the global market of 2.3 billion dollars. The color of the product imparts its pleasing aesthetics, sensory value, and quality. Natural colorants have emerged as an excellent substitute because of the significant health effects and undesirable demand for synthetic colorants. Because of awareness of the health benefits of natural colorants, its demands keep increasing with time. Several natural colorants are used in the food industries, out of which carotenoids, anthocyanins, and betanins were discussed, as they have broad applicability in the market. The research trend was finding the novel sources of natural additives and effective extraction methods of natural pigments. The various methods employed for extracting these pigments were enzyme-assisted extraction, supercritical fluid extraction, natural deep eutectic solvent, microwave-assisted extraction, ultrasound-assisted extraction, ionic assisted ultrasound extraction method. It was found that these novel extraction methods were highly effective and less time-consuming as compared to conventional extraction methods. Even after successful and effective extraction of natural pigments from various sources, the stability problem arises. Because of this, the degradation mechanisms of the natural pigment were discussed in brief. The natural pigments tend towards degradation at various pH levels, temperature, pressure, water content, and storage time. Various microencapsulation and nanoencapsulation techniques were discussed, which solves the stability problem of pigment.

CRediT authorship contribution statement

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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